Mathematics - Undergraduate Level Assessment 7 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If the finite group G contains a subgroup of order five but no element of G other than the identity is its own inverse, then the order of G could be

   (a) 8

   (b) 20

   (c) 30

   (d) 35

2. If A is a subset of the real line R and A contains each rational number, which of the following must be true?

   (a) If A is open, then A = R.

   (b) If A is closed, then A = R.

   (c) If A is uncountable, then A = R.

   (d) If A is uncountable, then A is open.

3. What is the volume of the solid in xyz-space bounded by the surfaces y = x^2, y = 2 - x^2, z = 0, and z = y + 3?

   (a) 8/3

   (b) 16/3

   (c) 32/3

   (d) 104/105

4. Let x_1 = 1 and x_(n+1) = sqrt(3+2 x_n) for all positive integers n. If it is assumed that {x_n} converges, then lim x_n =

   (a) 3

   (b) e

   (c) sqrt(5)

   (d) 0

5. Statement 1 | A ring homomorphism is one to one if and only if the kernel is {0}. Statement 2 | Q is an ideal in R.

   (a) True, True

   (b) False, False

   (c) True, False

   (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: Every ring must contain a multiplicative identity, which is not true for all rings.
7. True or False: Every group of order 42 has a normal subgroup of order 7 due to the Sylow theorems.
8. True or False: In the ring of continuous real-valued functions on [0, 1], the product of two nonzero functions can be zero.
9. True or False: If f is a linear transformation and f(1, 1) = 1 and f(-1, 0) = 2, then f(3, 5) equals 9.
10. True or False: The unity of a subring must be the same as the unity of the larger ring it belongs to.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. There exist two distinct unit vectors $\mathbf{v}$ such that the angle between $\mathbf{v}$ and $\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ is 45°, and the angle between $\mathbf{v}$ and $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ is 60°. Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be these vectors. Find $\|\mathbf{v}_1 - \mathbf{v}_2\|$.
12. Discuss the possible dimensions of the intersection of two 3-dimensional subspaces, U and V, within R^5. Analyze the conditions that affect these dimensions and provide examples to illustrate your reasoning.

End of Paper